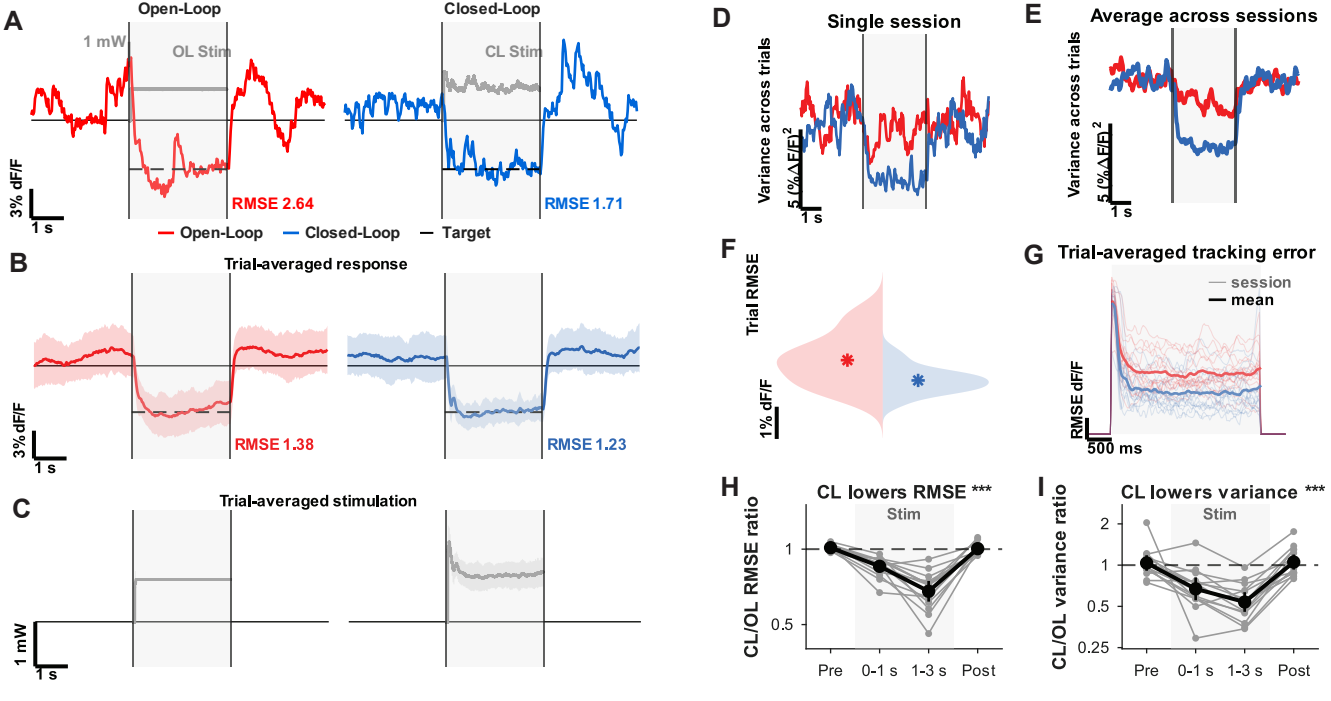

A
Open-Loop
Closed-Loop
1 mW
OL Stim
CL Stim
3% dF/F
1 s
RMSE 2.64
RMSE 1.71
Open-Loop
Closed-Loop
Target
B
Trial-averaged response
3% dF/F
1 s
RMSE 1.38
RMSE 1.23
C
Trial-averaged stimulation
1 mW
1 s
D
Single session
Variance across trials
5 (%ΔF/F)2
1 s
E
Average across sessions
Variance across trials
5 (%ΔF/F)2
1 s
F
Trial RMSE
1% dF/F
G
Trial-averaged tracking error
session
mean
RMSE dF/F
500 ms
H
CL lowers RMSE ***
Stim
CL/OL RMSE ratio
1
0.5
Pre
0-1 s
1-3 s
Post
I
CL lowers variance ***
Stim
CL/OL variance ratio
2
1
0.5
0.25
Pre
0-1 s
1-3 s
Post